

SAMBIT MISHRA

Ph.D. Student, Electrical & Computer Engineering | University of Southern California

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Research Interests

Causal inference and probabilistic graphical models, focusing on identifiability theory and scalable continuous-optimization methods for causal discovery, particularly efficient algorithms for identifiable DAGs on high-dimensional, mixed-distribution data.

Education

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Urbashi Mitra

GPA: 4.00/4.00

Los Angeles, CA, USA

Aug 2025 – Present

Key Coursework: Probability Theory, Linear Algebra, Supervised Machine Learning, Inference & Estimation Theory, High Dimensional Statistics and Learning Theory, Convex Optimization

Indian Institute of Technology (IIT) Bhubaneswar

B.Tech. (Hons.) in Electronics and Communication Engineering; Advisor: Dr. Soumya P. Dash

GPA: 9.14/10.00

Bhubaneswar, Odisha, India

2021 – 2025

Preprints & Publications

- [1] **S. Mishra** and U. Mitra, “On the Identifiability of Mixed Ordinal and Exponential Family Causal DAGs under Linear Parametric Models,” *arXiv preprint arXiv:2609.17942*, submitted to *Journal of Machine Learning Research*, 2026.
- [2] **S. Mishra** and U. Mitra, “Causal Discovery in Equal Variance Linear Gaussian DAGs via SURE-Tuned Ridge Regression,” *arXiv preprint arXiv:2608.17132*, accepted for oral presentation at 60th Asilomar Conference on Signals, Systems and Computers 2026
- [3] **S. Mishra**, Y. Wang, C. K. Johnson, and U. Mitra, “Epidemiological Causal Graph Identification: Challenges, Identifiability and Algorithms,” *arXiv preprint arXiv:2609.20676*, accepted for oral presentation at 60th Asilomar Conference on Signals, Systems and Computers 2026
- [4] C. Peng, **S. Mishra**, U. Mitra, “Learning to Intervene: Optimized Soft Intervention Selection for Causal Discovery,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, 2026, pp. 6196–6200.
- [5] **S. Mishra**, S. P. Dash and G. C. Alexandropoulos, “SER-Optimized Multi-Level ASK Modulations for RIS-Assisted Communications With Energy- and Sign-Based Noncoherent Reception,” in *IEEE Transactions on Green Communications and Networking*, vol. 10, pp. 1433–1445, 2025.
- [6] **S. Mishra** and S. P. Dash, “Error Analysis With Optimal Receiver and Multi-Level ASK for RIS-Assisted Noncoherent Wireless System,” in *IEEE Wireless Communications Letters*, vol. 15, pp. 300–304, 2025.

Research Experience

Graduate Research Assistant

Ming Hsieh Department of Electrical and Computer Engineering, USC

Los Angeles, CA, USA

Aug 2025 – Present

- Developed GPU-accelerated continuous optimization methods (PyTorch, Adam, L-BFGS) for scalable DAG estimation with differentiable constraints, reducing iterations and enabling large-scale batching.
- Designed a soft-intervention selection framework to cut experimental costs, and studied identifiability of causal graphs under finite samples and latent variables.

Undergraduate Researcher

School of Electrical and Computer Sciences, IIT Bhubaneswar

Bhubaneswar, Odisha, India

Aug 2024 – May 2025

- Developed SER-optimized modulation schemes for RIS-assisted noncoherent communication systems.
- Conducted error analysis for multi-level ASK modulations, optimizing energy efficiency.

Technical Skills

- **Causal Inference:** Causal Discovery, Bayesian Networks, Structural Equation Models.
- **Optimization:** Gradient-based solvers (Adam, RMSProp, L-BFGS), Convex Optimization, Bilevel Optimization.
- **Programming & Tools:** Python (PyTorch, NumPy, SciPy), C++, CUDA, MATLAB, Git, LaTeX.

Industry Experience

NVIDIA Corporation

ASIC Engineering Intern, CHI VIP Team

Bengaluru, Karnataka, India

May 2024 – July 2024

Verified CHI protocol compliance for CPU/GPU IPs using SystemVerilog and Perl automation; ensured interoperability between internal and external verification IPs.

Awards & Academic Distinctions

- B.Tech. Project Proficiency Award — Best B.Tech Thesis, ECE Department, IIT Bhubaneswar (2024–25)
- All India Rank 4000 of $\sim 1,000,000$ candidates, JEE Advanced 2021
- National Talent Search Scholarship — top 2000 of $\sim 1,200,000$ candidates, NTSE 2019, Government of India